Accessishelf: Increasing Safety Through Creating Automated Remote-Controlled Cabinetry

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Statement of Purpose

Over **110 million** Americans currently experience limited vertical mobility, including wheelchair users, the elderly, and vertically challenged individuals. This creates a critical need for accessible home cabinetry (USCB, 2024). These necessary accommodations are missing from current housing infrastructure, forcing users to rely on dangerous makeshift methods such as stepladders, chairs, or grabbers to access high storage spaces (CDC, 2023). These methods *significantly* increase the risk of **severe falls** and **injuries** in the home (CDC, 2023). As such, there is an existing and urgent demand for a retrofittable, automated mechanism capable of safely lowering storage compartments into the optimal ergonomic reach zone of 800-1100 mm without require excessive physical exertion from the user (Doe, 2025).

Background Research

The current accessibility challenge regarding home cabinets stems from the disconnect between standard architectural practices and the physical realities of mobility impaired persons. While the ADA caps the maximum reach for wheelchair users at 48 inches, the bottom of a standard U.S. kitchen cabinet typically begins at 54 inches; this 6-inch gap ensures that even the lowest shelf is inaccessible, with subsequent shelves drifting even further out of reach (Doe, 2025).

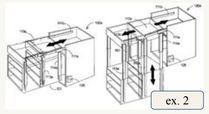
Because these storage spaces are out of reach, users are frequently forced to resort to unsafe alternatives. These can include climbing on countertops, balancing on dining chairs, or utilizing unstable step stools. This behavior directly contributes to domestic injury rates. Falls are the current leading cause of both fatal and nonfatal injuries among older adults and individuals with physical disabilities (CDC, 2023).

There have been attempts to address the accessibility gap, but existing products fail to provide a comprehensive solution.

- **Manual Pull-Down Systems (ex. 1):** Products like the Rev-A-Shelf system use gas struts or spring-loaded arms, but require significant upper body strength to operate, limiting accessibility for much of the target demographic (Rev-A-Shelf, 2024).
- **Automated Cabinetry (ex. 2):** Existing patented automated systems are functional but complex, costly, and require specialized installation, making them difficult to retrofit into standard residential cabinets (US Patent 8424983, 2013).
- **Remote-Controlled Shelf (ex. 3):** Products like the Automatic Lifting Shelf offer remote-controlled movement but typically only move vertically and are very expensive, leaving demand for a more affordable forward-and-downward solution (AliExpress, 2025).

This project's goal is to address inaccessible shelving by developing AccessiShelf, an automated and compact system that lowers and extends high shelves to improve accessibility. Some design goals include: supporting at least 15 pounds, fitting in standard cabinet dimensions, operating quickly and quietly, and plugging into a standard outlet.


ex. 1


ex. 2



ex. 3

Materials

These materials are specific to the team's final Generation 3 Prototype:

- Structural: Bathroom Wall Cabinet, ¾" Melamine White Particleboard, Motor Mounts
- Mechanical: Door Pivot Hinges, Pulley Wheels, Pillow Bearings, 8mm Axle Shaft, Winch Drum, Dyneema Rope, Door Closer
- Electronics: Neo V1.1 Brushless Motors, REV Spark Max ESCs, Raspberry Pi Zero 2W, Waveshare 2CH CAN HAT, 12v 100a Power Unit, Bluetooth Remote, 12v to 5v Micro USB Buck Converter, Jumper Cables, 120Ω Resistors, MicroSD Card

Design Limitations

The culmination of 3 generations of design produced a final product which accomplished all previous agreed upon design specifications, while leaving room for improvement. The presence of visible electronic components which required covering up is a major issue with the current team's design. Improvement in this area could be driven by increased focus on aesthetics, and custom electronic integration within the design. Furthermore, the team identified limits with the maximum movement speed of the motors. In commercial settings, this could present issues with customer satisfaction. As a solution, the team identified utilization of brushless DC motors which operate faster.

Prototype 1

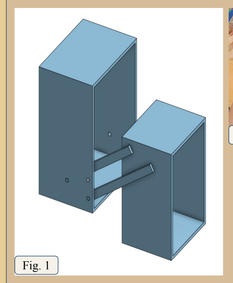

Fig. 1


Fig. 2


Fig. 2a

In the first prototype, Team AccessiShelf focused on demonstrating a proof of concept, showing that a working shelf could be made to pull down to users a sufficient distance. The device consisted of nested wooden shelves (Fig. 1) that are connected through a four bar mechanism and each maintain verticality and level throughout the motion. The prototype can extend up to 12" (Fig. 2) and return to initial positioning. An alternate view of the finished prototype is provided (Fig. 2a). The movement path was consistent and sufficient to demonstrate the idea of our design. Goals for Generation 2 include automation, further extension, and a support rope.

Prototype 2

The Generation 2 Prototype focused on the shelf's ability to lower to users while adopting a bent arm design (Fig. 3) that avoids stopping short of full extension, while still extending further. Shortcomings noticed in Generation 1 such as limited vertical mobility were corrected by changing the range of motion of the bars and positioning. The biggest change from Prototype 1 was the use of a remote controlled pulley and linear actuators; utilizing these allowed the user to pull the cabinet out remotely and control upwards-and-downwards movement (Fig. 4 & 5). As such, Prototype 2 demonstrated a proof-of-concept prototype in having a remote-controlled mechanism for the inner cabinet, which could support the inner cabinet's extended position (Fig. 6). Preliminary testing, which showed that the prototype could fully extend and support significant weight, is shown in the tables below. More work can be done to improve the smoothness of extension/retraction and weight bearing capacity, namely in the quality of material.

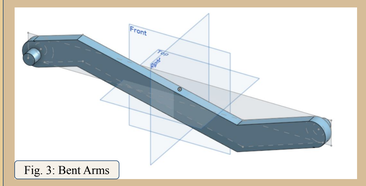


Fig. 3: Bent Arms

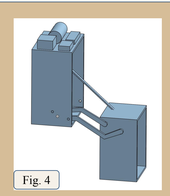
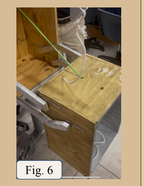

Fig. 4


Fig. 5


Fig. 6

Extension Distance and Mobility

Vertical Distance (in)	Comments on Stability
6.0	Level and aligned
12.0	Level, displaced slightly left
18.0	Slightly tilted backwards, aligned

Load Bearing Capacity and Strength

Motion tested	Weight tested (lbs)	Pass / Fail
Extension (down)	15.0	Pass
Bottom position	20.0	Pass
Retraction (up)	15.0	Pass

Prototype 3

In Generation 3, Team AccessiShelf focused on integrating the lifting mechanism with the cabinet housing while resolving aesthetic challenges from Generation 2. The final design incorporated two opening doors pushed outward by the cabinet structure (Fig. 7). After delays in refining the exterior, the team switched to a pre-made cabinet for improved aesthetics (Fig. 8) and began constructing the product around the newly made pulley system.

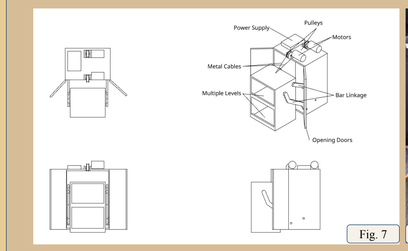
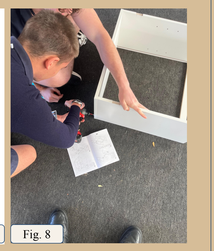


Fig. 7


Fig. 8

A larger focus was placed on the creation of a quieter pulley mechanism to lift the inner cabinet in an upwards-and-downwards motion. To accomplish this, the team utilized the pulley system seen above (Fig. 7), while designing code for the pulley system to be connected to the remote (Fig. 9) through an Arduino microcontroller. After connecting the pulley system to an external battery compatible with wall outlets and wirelessly connecting the remote to the pulleys, the team mounted the pulley system onto the pre-made cabinet, along with the battery source. Adjustment of the pre-made doors was necessary in order to ensure full functionality of the cabinet, as well as utilizing a new string-based cable to support the inner cabinet movement. The final prototype is shown in a kitchen setting (Fig. 10), and it subsequently was tested to ensure maximum utility.

```
import can
import struct
import time
import threading
import sys
import evdev
from evdev import ecodes

# --- CONFIGURATION ---
DEVICE_ID = 1

# Duty Cycle Command ID (Base + Device ID)
ARB_ID_DUTY_CYCLE = 0x02050080 | DEVICE_ID

# HEARTBEAT ID (Broadcast, no Device ID attached)
ARB_ID_HEARTBEAT  = 0x02052C80

# Speed (0.15 = 15% Power)
UP_POWER   =  0.15
DOWN_POWER = -0.15
STOP_POWER = 0.0

class WinchController:
```
Fig. 9


Fig. 10

Testing Results

The last part of the team's Generation 3 Prototype construction process was testing the prototype to ensure that it met all constraints previously established. Utilizing the testing procedures noted to the right, the team examined the maximum supportable weight, the stability of contained objects, the general auditory experience, and ease of remote use for the mechanism. Abiding by the procedures set earlier, the Generation 3 Prototype passed each test, with detailed observations of each testing process seen below in the tables. As such, the team completed the Generation 3 Prototype with verification that the prototype was reasonably quiet, strong, simple to use, stable, and significantly more aesthetically pleasing.

Remote Functionality Test

Cycle #	Extension Observations	Retraction Observations
1	Pass - Fully extended with no issue	Pass - Fully retracted with no issues
2	Pass - Fully extended with no issue	Pass - Stalled for a few seconds, but no need for manual intervention
3	Pass - Fully extended with no issue	Pass - Fully retracted with no issues

Stability of Internal Objects Test

Item number	Shift distance (mm)	Item standing (yes/no)
1	13	Yes
2	36	Yes
3	5	Yes

Maximum Weight & Sound Test

Weight (lbs)	Cycle complete (yes/no)	Structural observations	Audible observations (all below 65 dB)
5	Yes	Slight flexing at front edge of top board where front motor is mounted	Slight strain noise during transition point
10	Yes	Increased flexing at front board	Wood scraped a tiny bit
15	Yes	Increased flexing at front of top board	No notable noises

Expert Interviews



Mr. Donald Strum: President of Michael Graves
30+ years specializing in design/aesthetical engineering.

Mr. Marcos Riveras: Independent Contractor
Cabinetry Installation Expertise: Experienced with high-end cabinets.



Mr. Strum – After reviewing the automated cabinet, Mr. Strum advised prioritizing noise reduction, solving the mechanism for the exterior door to open during lifting, and ensuring the design fits standard cabinetry sizes for broader compatibility.

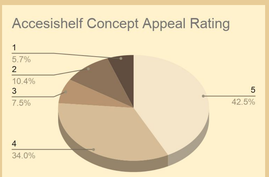
Mr. Riveras – Mr. Riveras, on the other hand, stressed the importance of aesthetics and material quality, recommending plywood over MDF and noting that cut edges require careful finishing to meet luxury cabinetry standards. He also emphasized that the design needs refinement to match high-end builds.

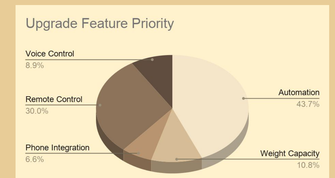
Market Research

Market research of 250 participants highlights a strong, yet unmet consumer demand for an accessible and retrofittable solution that bridges the gap between expensive custom builds and physically demanding manual shelves. The number of participants was chosen in context of the need to survey all parts of the market, with 250 being determined as a large enough number to reasonably include most interested demographics.

Consumer Demand & Pain Points: Survey results demonstrate that over 75% of respondents face daily challenges accessing their high cabinets, exhibiting the need for this product.

Willingness for Automation: More than 85% of survey respondents placed a high value on remote-control operation or an automated shelving system over existing manual alternatives. Responses typically cite independence and safety as their primary motivations for adoption.





Testing Procedures

Test 1: Maximum Weight Capacity Evaluation
Purpose: Confirm the Prototype 3 chassis can safely support a 15 lb payload
Pass/Fail: The system must complete a full extension and retraction cycle with no audible gear slippage, cracking, or crucial deformation.

Test 2: Payload Stability Assessment
Purpose: Ensure the mechanism's movement is smooth enough to prevent stored items from dangerously shifting
Pass/Fail: Test objects must remain standing and shift no more than 50 mm from their original placement after a full cycle.

Test 3: Remote Functionality & Integration
Purpose: Verify seamless hardware and software communication via a single Bluetooth remote
Pass/Fail: The shelf must successfully complete 3 consecutive cycles using only the remote, requiring absolutely no manual assistance.

Test 4: Acoustic & Sound Level Check
Purpose: Guarantee the lowering mechanism operates quietly enough for residential use
Pass/Fail: Peak noise levels must remain under 65 dB (measured from 1 meter away) throughout an entire descent and ascent cycle.

Conclusion

The AccesiShelf design successfully demonstrated the development of an automated accessibility solution through three generations of iterative design. While the team faced challenges in refining stability, automation, and integration throughout the process, the AccesiShelf evolved from a basic proof-of-concept lowering shelf to an automated customer-focused system. Generation 1 demonstrated the feasibility of a lowering shelf mechanism, while Generation 2 improved functionality by introducing a motorized winch and remote control to automate movement. Generation 3 addressed the resulting concerns of noise and bulk by prioritizing spatial efficiency, smoother motion, and better integration into existing cabinetry. The final design successfully met all primary design specifications and demonstrated the viability of an automated shelving solution for improved accessibility. Following Generation 3, potential future development could include the implementation of quieter motors, backup power options, and the use of lighter materials to further enhance system performance. AccessiShelf has proven it has the potential to make storage spaces more reachable for users with limited mobility.